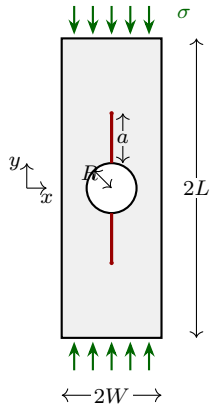
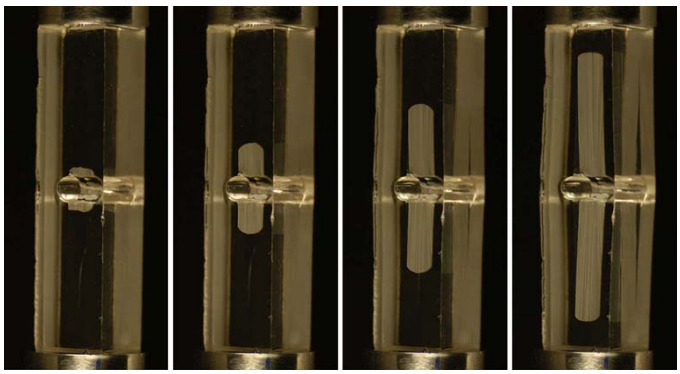
Fracture Mechanics 2026 · Homework

Véronique Lazarus (veronique.lazarus@ensta.fr) • Corrado Maurini (corrado.maurini@sorbonne-universite.fr)

Double Cleavage Drilled Compression test

We consider the *Double Cleavage Drilled Compression* (DCDC) test of the figures below, see *e.g.* [1-4].





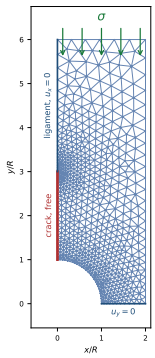


Figure 1. Geometry and loading; the thickness d is out of the plane. *Figure 2.* Experiments from [3], four successive instants. *Figure 3.* The quarter of the specimen that is meshed, at $a = 2R$ and $W = 2R$, drawn at $L = 6R$ so that the hole is visible.

Several authors propose to use this test for the experimental measure of the fracture toughness, see *e.g.* [3]. The specimen is compressed, but the two cracks open in mode I: the hoop stress around the hole is a tension at the two points on the axis of the load, and this tension drives them.

The aim of this homework is to calculate the energy release rate of the crack as a function of its length, using different methods, and to predict the evolution of the crack as a function of the loading.

The use of numerical solutions through the finite element method is strongly appreciated. However, students who have difficulties with the installation of the required software and with programming can proceed using the approximate formulas provided in the literature. A minimal list of references is given below, but other references can be used, with appropriate acknowledgment of the sources.

In the whole work we assume small deformations and a linear elastic, isotropic and homogeneous material, with Young's modulus E and Poisson ratio ν. If not otherwise specified, $\nu = 0.2$ and we work in **plane strain**. The structure is loaded by the uniform compressive stress σ shown in figure 1. Without loss of generality, we perform the numerical simulations with $E = 1$, $\sigma = 1$ and a hole radius $R = 1$, see Question 1 below.

Each group has its own pair $(W/R,\ R)$**.** You find it on the parameter sheet given with this subject. Questions 3, 4 and 5 depend on it; everything else is common.

Two points cost a factor on G if you miss them, so they are worth stating once.

- **Take a long specimen,** $L \geq R + a + 3W$. If the loaded face is closer to the crack tip, it changes the stress there and none of the published formulas applies. The scripts use this value by default.
- **The mesh covers one quarter of the specimen, and the specimen has two crack tips.** The potential energy of the whole specimen is four times the computed one, and the energy release rate of one tip is $G = -\frac{1}{2}\,\mathrm{d}\Pi/\mathrm{d}a$.

Questions

1. Suppose that, all the other parameters being fixed, you perform a numerical simulation with $E = 1$, $\sigma = 1$ and $R = 1$. Let us denote the resulting displacement field by $\underline{u}^*$, the strain field by $\underline{\underline{\varepsilon}}^*$, the stress field by $\underline{\underline{\sigma}}^*$, the energy release rate by G^*, and the stress intensity factor by K_I^*. Give, by dimensional analysis and linearity of the elastic problem, the expressions of $\underline{u}$, $\underline{\underline{\varepsilon}}$, $\underline{\underline{\sigma}}$, G and K_I for any value of E, σ and R. Summarise the result in a set of equations. Deduce that $\hat{g} = G\,E/(\sigma^2 R)$ depends only on a/R, W/R, L/R and ν.

2. Compute the stress intensity factor for $a = 2R$, with $W = 2R$ and a long specimen, using the different methods below. Perform the calculations for different mesh sizes and discuss the influence of the mesh on the results. Summarise all the results in one or two tables and comment.
   - The crack opening displacement method

- The numerical derivative of the potential energy
- The G–θ method

3. Compute the energy release rate for $a \in [0.01,\, 2]\, R$, with the value of W/R of your group. Try to find in the literature an approximation of the ERR (or of the SIF) which is valid in this regime, and compare it with your numerical results. Ideally, compare the results obtained with two different numerical methods (G–θ and the finite difference of the potential energy). Summarise all the results in a single figure and comment. In particular: where is $\hat{g}$ maximum, and what does this say about a small flaw left by the drilling?

4. Compute the energy release rate for $a \in [2,\, 10]\, R$, with the value of W/R of your group. Same comparison with the literature and between two numerical methods. Summarise all the results in a single figure and comment. In particular: on this branch $\mathrm{d}G/\mathrm{d}a < 0$, so what happens to a crack that grows at a fixed σ, and what does this test give that a three-point bending test does not give?

5. Assume that the plate is made of concrete, with the hole radius R of your group. List the corresponding typical values of E, σ_c and G_c, and give your source. Structural safety is assured if $\sigma \le \sigma_c$ and $G(a) \le G_c$. Plot the maximum allowable load σ versus the crack length and comment. Compare the material length $\ell_{\mathrm{mat}} = EG_c/\sigma_c^2$ with the radius R of your group: can the test be run on a specimen of this size?

Bonus. Can you conceive of a small experimental test to estimate the fracture toughness of emmental cheese (or a similar material)? Do it!

The report

- One pdf per group of two, **4 pages maximum**, plus the cover sheet, which does not count.
- Write your group number and your pair (W/R, R) at the top.
- Label the axes of your figures: the quantity and its unit, or the dimensionless group when there is no unit.
- Name the file `MauriniStudentName1StudentName2.pdf` (French class with C. Maurini) or `LazarusStudentName1StudentName2.pdf` (English class with V. Lazarus).
- Submission folder and deadline: *to be announced*.

AI tools

You may use AI tools, or not. If you do, write on the cover sheet which tool you used and for what, question by question. This declaration is part of the mark. A short report that you understand is better than a complete report that you cannot explain.

Individual check

At the last lecture: ten multiple-choice questions, ten minutes, each student alone, with their own report and nothing else. There is no new physics in it. Each question asks about a decision you made while doing the homework, or about your own curves; the sheet checks that you did the work and that you know what your report says. Each question has one correct answer, worth +1; any other answer is worth 0, and nothing is taken off, so answer all ten.

Five correct answers out of ten, that is 10/20, validate the homework. Your final mark is then $\max(0.2\,\text{homework} + 0.8\,\text{exam},\ \text{exam})$. Without the validation it is the exam mark alone. So the homework can raise your mark, and it can never lower it.

References

[1] Nielsen, Amirkhizi & Nemat-Nasser (2014). An empirical model for estimating fracture toughness using the DCDC geometry. *Int. J. Fract.* 188, 113–118.

[2] Pallares *et al.* (2009). Crack opening profile in DCDC specimen. *Int. J. Fract.* 156, 11–20.

[3] Plaisted, Amirkhizi & Nemat-Nasser (2006). Compression-induced axial crack propagation in DCDC polymer samples. *Int. J. Fract.* 141, 447–457.

[4] Fett & Rizzi (2005). *A fracture mechanics analysis of the DCDC specimen*, FZKA 7094, Forschungszentrum Karlsruhe.

The fits write $K_I = \sigma\sqrt{\pi R}\, F(a/R,\, W/R)$ and give $1/F$ as a straight line in the crack length, $\sigma\sqrt{\pi R}/K_I = A + B\, a/R$. The second term uses a/R, and not a/W.

Practical notes

Nothing here is asked for in the report. These are the points on which the computation usually goes wrong.

- G–θ**.** Choose the inner and the outer radius of the ramp, then change both and see how much G moves. The ramp must not touch the hole, the free side or the loaded face, so choose its size from $\min(a,\, W,\, L-R-a)$ and not from a alone.
- **Finite difference of the potential energy.** A large step Δa gives a bias, a small step gives noise. Two crack lengths need two meshes, so the difference also contains the noise of the meshes.
- **Crack opening.** The crack opening is twice the displacement computed on the symmetry plane. Draw the apparent $K_I(r)$ and extrapolate it to $r = 0$: the small r are spoiled by the size of the elements and the large r by the higher order terms. This method converges more slowly than the other two.
- **The mesh.** It is graded: small elements at the crack tip, large elements far from it. Study the convergence with the size of the elements near the tip. 25 to 30 crack lengths, spaced logarithmically below $a = R$, are enough for the curves of Questions 3 and 4, and the whole curve takes a few minutes on a normal laptop.
- **A cheap check for Question 3.** A crack much shorter than R sees the hoop stress of the hole as uniform over its own length, so it behaves like an edge crack in a half plane, for which $K_I = 1.1215\, s\sqrt{\pi a}$. Drawing this straight line on your figure finds an error of a factor 2 or 4 at once.
- **Question 5.** Check that the values of G_c and K_{Ic} you take from the literature satisfy $G_c = K_{Ic}^2/E'$, with $E' = E/(1-\nu^2)$ in plane strain. Handbooks give one or the other, and the two are not always consistent.